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PROFILE

- MSc by Research student at the Institute for Computing Systems Architecture (ICSA), University of Edinburgh, supervised by Prof. Mahesh K. Marina. My work sits at the intersection of **machine learning systems** and **networked systems**: how a training or inference workload shares an accelerator and a network with a latency-critical tenant, and how such systems are measured at a scale no physical testbed can reach.

Two lines run through everything I have built. The first is **resource sharing under a hard deadline** — deciding how much GPU capacity an AI workload may harvest from infrastructure it does not own, on a control-loop timescale set by the priority tenant, and keeping the harvesting workload’s state intact when that capacity is reclaimed. The second is the **high-performance data plane** underneath: I own the C++ inter-device transport in *Morphling* (event-driven I/O, zero-copy scatter-gather, pinned-memory pooling, serialization-cost analysis), where end-to-end fidelity depends on microsecond-level transfer behaviour. This work has produced papers at ACM MobiCom and a Best Paper award at EdgeSys’26.

I am applying for the PhD programme at iSING Lab for Fall 2027 entry, and intend to apply to the Hong Kong PhD Fellowship Scheme. What draws me to the lab is the **AI-centric networking** and **machine learning systems** directions: I have worked on the compute side of AI infrastructure sharing and on the transport layer that carries it, and I want to push the same questions down into the datacenter network itself — RDMA, SmartNICs and programmable switches — rather than treating the network as a given.

EDUCATION

- **University of Edinburgh — MSc by Research, Computer Science** Edinburgh, UK
Supervised by Prof. Mahesh K. Marina. Thesis: Flux-DT: An Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN Sep 2025 – Sep 2026 (expected)
 - Research-only degree (thesis, no taught coursework); sole author of the thesis and its implementation.
- **University of Edinburgh — BSc (Hons) Computer Science** Edinburgh, UK
*Grade: **First Class Honours** (weighted average 74/100; UK First Class $\geq$ 70/100)* Sep 2021 – May 2025
 - **70+** in *Machine Learning Systems*, *Machine Learning Practical*, and *Computer Communications and Networks*.

PUBLICATIONS AND MANUSCRIPTS

EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System *ACM MobiCom’26 (cond. accepted)*
Ujjwal Pawar, Andrew E. Ferguson, Yufeng Xia, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic
My role: owned the end-to-end evaluation — cloud deployment at 130 instances / 6,400 cores, fidelity and scalability campaign.

Morphling: Emulator for Distributed Machine Learning at the Edge *EdgeSys’26 — Best Paper Award*
Leyang Xue, Yufeng Xia, Eren Mendi, Ismaeel Bashir, Jiaxun Yang, Myungjin Lee, Mahesh Marina
My role: second author; owned the C++ inter-device transport data plane and the distributed virtual-clock synchronisation.

Weaver: Foundation Model Training over AI-RAN Compute Infrastructure *under submission*
Leyang Xue, Tianxin Wang, Xin Zhe Khooi, Jiaxun Yang, Dheeraj Mahendiran, Yufeng Xia, Mun Choon Chan, Myungjin Lee, Mahesh Marina
My role: GPU profiling and PHY-layer interference characterisation; controller evaluation and large-scale training runs.

Wasp: Foundation Model Training System for the Edge *ACM/IEEE SEC’26*
Leyang Xue, Yufeng Xia, Myungjin Lee, Mahesh Marina
My role: second author; TODO-CONFIRM-WASP-ROLE.

CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization *under submission*
Yufeng Xia, Pei Mu, Antonio Barbalace
My role: **first author** — problem formulation, LLVM implementation, evaluation and writing (manuscript attached).

- **Weaver — Foundation Model Training over Shared AI Compute Infrastructure** Under Submission
University of Edinburgh — GPU profiling, interference characterisation, controller evaluation 2024 – 2026
 - **Harvesting accelerators from a latency-critical tenant:** Weaver lets foundation-model training run on GPUs that a latency-critical workload owns, harvesting up to **83%** of the available spare compute on a multi-site testbed without degrading the priority tenant. The core question — how much capacity may be taken, and how fast the decision must be revised — is a near-real-time resource-allocation problem on shared accelerators.
 - **Why the sharing works: a roofline argument:** Ran the GPU profiling and roofline analysis showing the priority workload’s kernels are memory-bandwidth-bound (arithmetic intensity two to three orders of magnitude below the device ridge point), while training is compute-bound — the asymmetry that makes co-location possible at the SM level rather than merely time-slicing.
 - **Deadline-aware measurement:** Instrumented the priority workload’s scheduler to measure per-deadline completion under co-location, and carried out the large-scale distributed training runs used in the evaluation across a multi-site deployment with a measured inter-site bandwidth/latency matrix.
- **Morphling — Emulator for Distributed Machine Learning at the Edge** EdgeSys’26, **Best Paper**
University of Edinburgh — owned the C++ transport data plane Oct 2025 – Mar 2026
 - **Inter-device transport:** Built the event-driven (libevent) proxy client/server that carries tensor traffic between emulated devices — roughly **2,400 lines of C++** across `proxy_svr`, `proxy_cli` and the connection layer, in an 18k-line C++/CUDA codebase. Emulation fidelity is bounded by this path, so transfer cost had to be driven down to the microsecond level.
 - **Zero-copy and memory-path optimisation:** Designed a bucketed `cudaHostAlloc` pinned-memory pool and zero-copy scatter-gather buffers to remove per-transfer allocation and staging copies; added CUDA host-registration and worker-thread pinning, each validated with dedicated microbenchmarks.
 - **Serialisation and batching cost analysis:** Measured serialisation overhead across Protobuf and FlatBuffers along the full send path (serialise / size / transmit), and used the results to choose the batching and framing strategy; wrote up the analysis as design documents in the repository.
 - **Distributed correctness:** Implemented virtual-clock synchronisation so that event-driven virtual time stays consistent across emulated devices, and diagnosed and fixed the connection-cleanup and device-failure deadlocks that appeared at scale.
- **Flux-DT — Elastic, Preemption-Tolerant Runtime for Stateful Workloads on Reclaimable Compute** MSc by Research Thesis
University of Edinburgh — sole author: problem formulation, design, implementation, evaluation Sep 2025 – Present
 - **State that must survive preemption:** Existing compute-harvesting designs assume the harvesting workload is stateless and disposable, so reclaiming its capacity means killing it. Flux-DT targets a workload that is neither: it carries scene state that must survive interruption. The runtime lets it run on bursty, reclaimable GPU capacity without violating the priority tenant’s guarantees or corrupting its own state.
 - **Graded response instead of eviction:** Designed and implemented a control loop that answers resource pressure with an ordered set of actions — relocate, reduce fidelity within bounds the priority workload tolerates, then suspend and resume — on the timescale the priority tenant imposes, rather than a single all-or-nothing eviction decision.
 - **Placement heuristic and shedding order:** Formulated work placement as a joint objective across mutually interfering units and designed a **greedy heuristic** that decides which work is worth doing under interference; the same ranking determines what is shed first when compute is reclaimed mid-solve, because the decision must return faster than an exact solver could.
- **EmuRAN — Scalable, Cost-Effective Cloud-based Emulation System** ACM MobiCom’26
University of Edinburgh + Microsoft Research — owned the end-to-end evaluation Conditionally Accepted
 - **Operating an emulator at cloud scale:** Deployed and operated the system on public cloud across **130 instances and 6,400 CPU cores**, emulating 10,000 mobile devices and 10,000 base stations — two orders of magnitude beyond existing software solutions.

- **Fidelity, not just scale:** Designed and ran the evaluation campaign: fidelity against real hardware (contention-based random-access timelines, iperf3 throughput comparison, per-slot completion-time distributions) and scalability (time-dilation behaviour against node count and under constrained compute), establishing that the system behaves correctly rather than merely running large.
- **CoSenseAQ — LLVM Auto-Quantization from Sensor Specifications** Under Submission
*University of Edinburgh, ICISA — **first author**, supervised by Dr. Antonio Barbalace* Apr 2024 – May 2025
 - **Compiler-level optimisation from out-of-band information:** Built an LLVM auto-quantization framework driven by sensor value ranges that a compiler cannot recover from source alone, performing variable-range analysis and type shrinkage to reduce code size and execution time on embedded targets.
 - **Measurement on real hardware:** Evaluated on ARM Cortex-M0 and Cortex-A53, measuring power, performance and area; low-level C/C++ and cross-compilation toolchain work throughout. Drove the work from problem formulation to manuscript.

TECHNICAL SKILLS

- **Languages** C++ (primary; C++17, the language of all my systems work), C, Python, Java, Bash.
- **High-Performance Data Plane** Event-driven asynchronous I/O (libevent), zero-copy scatter-gather buffers, pinned-memory pooling and CUDA host registration, serialisation cost analysis (Protobuf, FlatBuffers), batching and framing, connection lifecycle management, thread pinning, microsecond-level latency instrumentation.
- **GPU and ML Systems** CUDA programming and profiling (Nsight), SM-level partitioning via green contexts and MPS, cuBLAS/cuBLASXt, roofline and arithmetic-intensity analysis, PyTorch, distributed foundation-model training, scheduling under contention with a latency-critical co-tenant.
- **Distributed and Cloud** Multi-site deployment on Microsoft Azure and CloudLab (130+ instances, 6,400 cores), Kubernetes, Docker, distributed virtual-time synchronisation, failure and deadlock diagnosis at scale.
- **Compilers** LLVM (IR-level passes, variable-range analysis, automatic quantization), MLIR (own multi-target tensor compiler with operator fusion, NVVM/ROCDL lowering to PTX/cubin), cross-compilation for embedded ARM.
- **Networking and Wireless** 5G RAN experimentation with OpenAirInterface and Open5GS, RAN emulation and network digital twins, throughput/latency measurement (iperf3, packet-level tracing).
- **Engineering** Git and code review, CMake, GitHub Actions CI, pre-commit, CodeQL, containerised reproducible builds, technical documentation (see attached toolchain and repository summaries).

INDUSTRY EXPERIENCE

- **Senior Tech — Back-end Development Intern** Suzhou, China
Production backend APIs in a commercial engineering team; live platform migration from JFinal to Spring Boot, code review and release testing. Java, MySQL, Redis, Kafka. Jun 2023 – Sep 2023

CODE

- **Morphling** github.com/netsys-edinburgh/morphling — EdgeSys'26 Best Paper artifact, 18k lines of C++/CUDA. I own the inter-device transport data plane (91 commits). *Detailed in the attached repository summary.*
- **TensorOps** github.com/xyf2002/tensor-compiler — sole-authored multi-target MLIR tensor compiler with operator fusion, lowering to NVVM/ROCDL and PTX/cubin execution via the CUDA Driver API.

AWARDS, TEST SCORES AND SERVICE

- **Award Best Paper**, EdgeSys 2026 (9th Workshop on Edge Systems, Analytics and Networking, co-located with ACM MobiSys 2026), for *Morphling*.
- **Test Scores** **GRE 336/340** (Verbal 166, Quantitative 170/170, Jun 2024); IELTS 7.0.
- **Service** Student volunteer, EuroSys 2026 (on-site conference operations).
- **Languages** Mandarin (native), English (fluent).

AVAILABILITY

- MSc by Research thesis submits in **September 2026**. I can join research projects **remotely from September 2026**, and am available **full-time from October 2026** — remotely or on site in Hong Kong — as a Research Assistant. I am applying for **PhD entry in Fall 2027** and intend to apply to the **Hong Kong PhD Fellowship Scheme**.